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ORIGINAL ARTICLE

Rural electrification in Kenya: a useful case for remote areas in sub-Saharan Africa

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Abstract There is a significant proportion of the world's population living in remote rural areas that are geographically isolated and sparsely populated. This study is based on modeling, computer simulation, and optimization of a hybrid powered mini-grid for a remote area of Korr in the district of Marsabit, Northern Kenya. The solar photovoltaic and wind turbine are considered as the two renewable resources for generating electricity accompanied by a battery (B) for storage and a diesel generator as a backup system. HOMER Pro software is used to perform the design and analysis of a proposed hybrid powered mini-grid model. The simulation results generated by the software indicate that the renewable energy sources may be a competitive technology. It has the potential of being a feasible solution capable of providing sustainable and reliable electric power at remote locations provided adequate amounts of renewable energy resources. The most cost-effective system in this study incorporates photovoltaic arrays, wind turbines, a diesel generator, and a battery bank to provide electricity to the community load demand of 5592 kWh/day at a cost of energy of $0.314.

Keywords Rural electrification · Kenya · Off-grid power system · Mini-grid

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Introduction

Electricity is a vital commodity that has proven to be a blessing bestowed on mankind from science. Unfortunately, for an estimated 1.2 billion people around the world their reality does not involve electricity meaning they lack access to electricity. This accounts for 16% of the world's population. In sub-Sahara Africa, 620 million people lack access to electricity which accounts for more than half of the globe's total population lacking access (IEA 2016). If someone adds to this, that Africa is the second largest refugee-hosting area in the world (Malekpoor et al. 2018), the problem of electricity access becomes worse. According to the International Energy Agency, sub-Sahara Africa will require more the US$979 billion in investment to achieve universal electricity access by 2030 (World Energy Outlook 2012 2012). Diesel generators are the predominant form of energy in rural areas in sub-Sahara and in particularly Kenya. These account for 83% of the total electricity that provides lighting to rural communities in Kenya (Energy Research Centre of the Netherlands 2015).

Kenya has a vision which aims at creating "a globally competitive and prosperous country with a high quality of life by 2030" (Kenya Vision 2030 2007). The aim is to make Kenya a newly industrialized, middle income country providing a high quality of life to all its citizens into a clean and secure environment. To achieve this, Kenya must prioritize the energy sector as it has huge potential. Households in Kenya still use kerosene as the main source of lighting in both rural and urban areas. It currently costs approximately US$339 to connect to the national grid and about US$0.19 per kWh of electricity service. These costs are high and pose a major obstacle for the expansion of electricity connections to low-income households and small business which may be able to benefit from alternative sources of energy such as wind and solar. The IEA forecasts that mini-grids, stand-alone off-grid solutions (Michalitsakos et al. 2017) will be necessary for 70% of all rural areas in developing countries (IEA 2016).

There are different schemes implemented to electrify rural areas in developing countries. They can be classified into three main categories: (a) grid extension, (b) mini-grids, and (c) stand-alone power systems (Mandelli et al. 2016; Vieira et al. 2017). These differ from each other in terms of project size, feasibility, effort and complexity. These brings along environmental (Panagiotidou et al. 2016) and social impacts (Hills et al. 2018; Weir 2018) and show various benefits as well as drawbacks (Nerini et al. 2016).

Most countries prefer the grid extension approach as the method of electrification which can be costly. As can be seen in Table 1, the costs may vary significantly depending on the country where the grid extension is occurring. However, from the perspective of potential customers, this method does not necessarily mean that electricity supply will be reliable or be able to meet the energy demands of a rural community.

Only a few publications have analyzed the cost of renewable mini-grids. A recent study from sub-Sahara found that a solar PV-diesel mini-grid that supplies 100 users can cost from US$0.46/kWh to US$0.74/kWh, while a 100% solar PV mini-grid would cost US$0.467/kWh to US$0.714/kWh (International Renewable Energy Agency 2016a). Stand-alone systems can be categorized into pico, home, and productive systems (Conway et al. 2013). It is estimated that 26 million households or else 100 million people worldwide are served through these off-grid mini-grids and stand-alone renewable energy based systems (REN21 2016).

As of 2015, there are approximately 400 GW of diesel capacity in operation either in the form of industrial purposes and mines operating remotely as backup units for unreliable electricity supply or as a community grid. The characteristics of diesel generators are shown in Table 2. Africa has 35 GW of diesel generated capacity (International Renewable Energy Agency 2016a).

As of 2015, it is estimated that more than 6 million solar home systems (SHS) and kits are operational worldwide, providing electricity for radio, television, refrigeration, lighting, and internet access (REN21 2016). This is gaining more interest—in an unclear and not organized way especially over the last few years. Seventy countries worldwide have off-grid solar capacity installed or programs in place to support off-grid solar application, as the end of 2015. In the period of 2014–2015, M-KOPA Solar sold approximately 300,000 SHS in Kenya, Tanzania, and Uganda (REN21 2016). In Table 3, the basic characteristics (most of them retrieved from HOMER's library) of the most common sub-systems that comprise a hybrid energy system (HES) are shown.

RES hybrid system technology offers a suitable alternative to single technology-based system for electrification by offering high-energy efficiency that is a result of combining more than one renewable energy technology (Chauhan and Saini 2014). Such systems have the capability of integrating the advantages of each technology and balance their different limitations (Mohammed et al. 2014). Energy efficient off-grid appliances occupy more and more space in sub-Saharan countries. The usage of non-electric models is promoted (solar ovens, composting toilets, etc.), and this could

Table 1 Costs of extending the grid per kilometer (Alliance for Rural Electrification 2014)

Country	Labor and other costs	Materials and products	Total
Mali	US$2590	US$15,170	US$17,760
Senegal	US$5150	US$10,810	US$15960
Kenya	US$6590	US$5960	US$12,550
El Salvador	US$2090	US$6160	US$8250
Laos	US$1420	US$7230	US$8650
Bangladesh	US$350	US$6340	US$6690

Table 2 Characteristics of diesel generators (Parajuli et al. 2015; HOMER Energy 2017)

Indicators	Diesel generators
Installed cost (US$/W)	0.4–0.9
Operations and maintenance (O&M) costs	1–10% of investment cost/year
Efficiency (%)	27–40
Lifetime (years)	Approx. 10*
Fuel cost (US$/lt)	0.91

*If diesel generator has a lifetime of 20,000 operating hours and HOMER determines that it operates 2000 h per year, its expected lifetime in years is 10 years

Table 3 Characteristics of PV, wind, hydropower, biomass systems (International Renewable Energy Agency 2016a; Ebhota and Inambao 2017; HOMER Energy 2017)

	PV systems	
	Small-scale (roof top)	Large-scale (utility scale)
Installed cost (US$/W)	2.00–3.50	1.0–2.5
Levelized cost of energy (LCOE) (US$/kWh)	0.15–0.25	0.10–0.20
Lifetime (years)	25–30	
O&M (US$/kW/year)	1.5% of installed cost	
	Wind systems (WS)	
	Small wind (hor. axis)	Large wind (hor. axis)
Installed cost US$/W)	4–4.5	1.5–3.5
LCOE (US$/kWh)	0.12–0.28	0.06–0.14
Lifetime (years)	>20–25	
O&M (US$/kW/year)	50–80	
	Hydropower (H)	
	Pico, micro-, and mini-hydro	Small hydro
Installed cost US$/W)	3.4–10	1.0–4.0
LCOE (US$/kWh)	>0.27	0.02–0.19
Lifetime (years)	30	>30
O&M (US$/kW/year)	45–250	
	Small-scale biomass gasifiers (BIO)	
Installed cost (US$/W)	2.0–5.0	
LCOE (US$/kWh)	0.07–0.24	
Lifetime (years)	5–12	
O&M (US$/kW/year)	5–6.5% of installed cost	

even influence the load peaks and the levels of integration of renewable energy sources (Khan et al. 2016).

Several researchers have conducted case studies on hybrid systems across the globe—and not only in developing countries—that provide energy to villages in rural areas. Bekele and Boneya (2012) studied a hybrid electric power generation system in Ethiopia consisting of a diesel generator (D), a PV, a wind power system (WS), and a battery unit (B). Rehman and Al-Hadhrami (2010) worked on a D + PV + B system for Rawdat Ben Habbas village, Saudi Arabia, where the annual CO_2 emissions were reduced by 4976.8 tons, saving 10,824 barrels of fuel per year. Another HES (PV + B + D) in Kolkata, India reduced the fuel costs by 70–80% compared to the diesel-alone system and reduced the CO2 emissions and other harmful gases by 90%. It was also the most cost-effective solution for the location with 95% renewable fraction (Nema and Dutta 2012). The feasibility of a wind/solar photovoltaic/diesel generator-based hybrid power system in a remote location in Fiji islands was also studied. The fully renewable-based system was more feasible than the conventional one, which had a 10% annual capacity shortage (Lal and Raturi 2012). A DER-CAM and storage-based analysis of a HES for a Greek island using mixed integer linear programming proved that 7 out of 11 proposed solutions were found economically feasible and beneficial for all stakeholders (Michalitsakos et al. 2017). Chauhan and Saini (2015) studied a multi resources HES utilizing all the available technologies such as BIO + H + PV + WS + B (where “H” is hydropower), depending each time on the availability of the resources in Uttarakhand, India.

The reason behind this study was not to simulate and present the results for simply another HES study. Knowing that the RE costs—especially for solar—are day after day going down, we took into account that even at the time of simulations and writing of this paper the costs would have already decreased further. The multi-objective optimization based on the variability of all the available resources in a low-income rural area in Africa taking into account the specific load peaks was one of the major scopes of this work.

The scope of this study was limited to determining the best cost-effective combination of electricity generation technology to the electrification of a selected area in Kenya. In addition, the evaluation for performance of the system was conducted.

Methodology

A deductive approach was taken in this study as it complements the research paradigm and with the sole purpose of identifying the most cost-effective system. However, an inductive approach was necessary when sizing the loads of the village due to lack of data. A mixed method was followed, where mostly quantitative method was used in the literature review while a qualitative method used in the case-study section. Quantitative and qualitative research methods are set into motion to ensure triangulation of the collected data.

Several assumptions regarding data acquisition were taken:

- Wind speeds and solar energy data from NASA Surface Metrology were contemplated as accurate for computing and simulating off-grid electrification systems.
- The rate of inflation will be assumed the same for all types of costs (operating and maintenance cost, labor cost, fuel costs, and so on) occurring over the 20 years.
- An approach to get the most accurate costs of the power system components and installations costs was attempted.

The main elements of this study are shown in the framework as the analysis of initial site assessment, system design, and techno-economic analysis resulting with the most cost-effective combination of electricity generation technology in an off-grid system for electrifying the selected area in Kenya.

The framework (Fig. 1) illustrates how, in different contexts, the most cost-effective combination of renewable energy resources (Koroneos et al. 2012) is realized by modeling and simulating by using the HOMER software. It is possible to model and simulate different systems by combining various input data such as the load profile, renewable energy resources, and the cost per kilowatt for the specific renewable energy technologies.

Kenya as a case study

As of 2016, Kenya has an estimated population of 46.7 million with 25.6% of this population in urban areas. The country has a severe problem with a rural electrification rate of only 7%, which is one of the lowest in sub-Sahara Africa (IEA 2016). Kenya added 1.3 million households to its electricity grid in 2016. This has seen the national electrification rate go from just 27% in 2013 to 55% as of the end of June 2016. The country's electrification rate is one of the fastest in the region. Electricity demand has steadily been increasing in Kenya with an electricity demand of 1586 MW for 2015–2016 compared to that of 1194 MW for 2010–2011 (Kenya Power Lighting Company 2016).

However, the electricity consumption per capita of 171 kWh which was last measured in 2014 was above the average of the region at the time but well below developed countries such as Denmark with an electricity consumption of 5859 kWh per capita or the global average of 3144 kWh per capita (The World Bank 2014). The predominant reason for such a low consumption per capita and low consumption overall can be related to the access to electricity outside densely populated urban areas remains low.

Site details

Korr is a remote village that has been selected to conduct the analysis for the various combinations of renewable energy system for supplying electricity. Korr is in northern Kenya in the county of Marsabit. The main economic activity is livestock hearing and commodity trading with both wholesale and retail taking place (Energy Research Centre of the Netherlands 2015). The terrain around the village is rocky and stony with the underground water level being very low. Figure 2 gives an aerial view of the location. It is located 55 km northwest of the Laisamis off-grid station and 63 km approximately southwest of Marsabit KPLC mini-grid. Both these systems generate electricity through diesel generators.

SHS are popular among the locals; however, the primary source of energy used for lighting by households is fuelwood, accounting for 76.2%. There is no electricity service from the national grid or a public or private mini-grid. No vehicle batteries, users with their own or shared generator, or other electrical energy sources were found among the end-user sample.

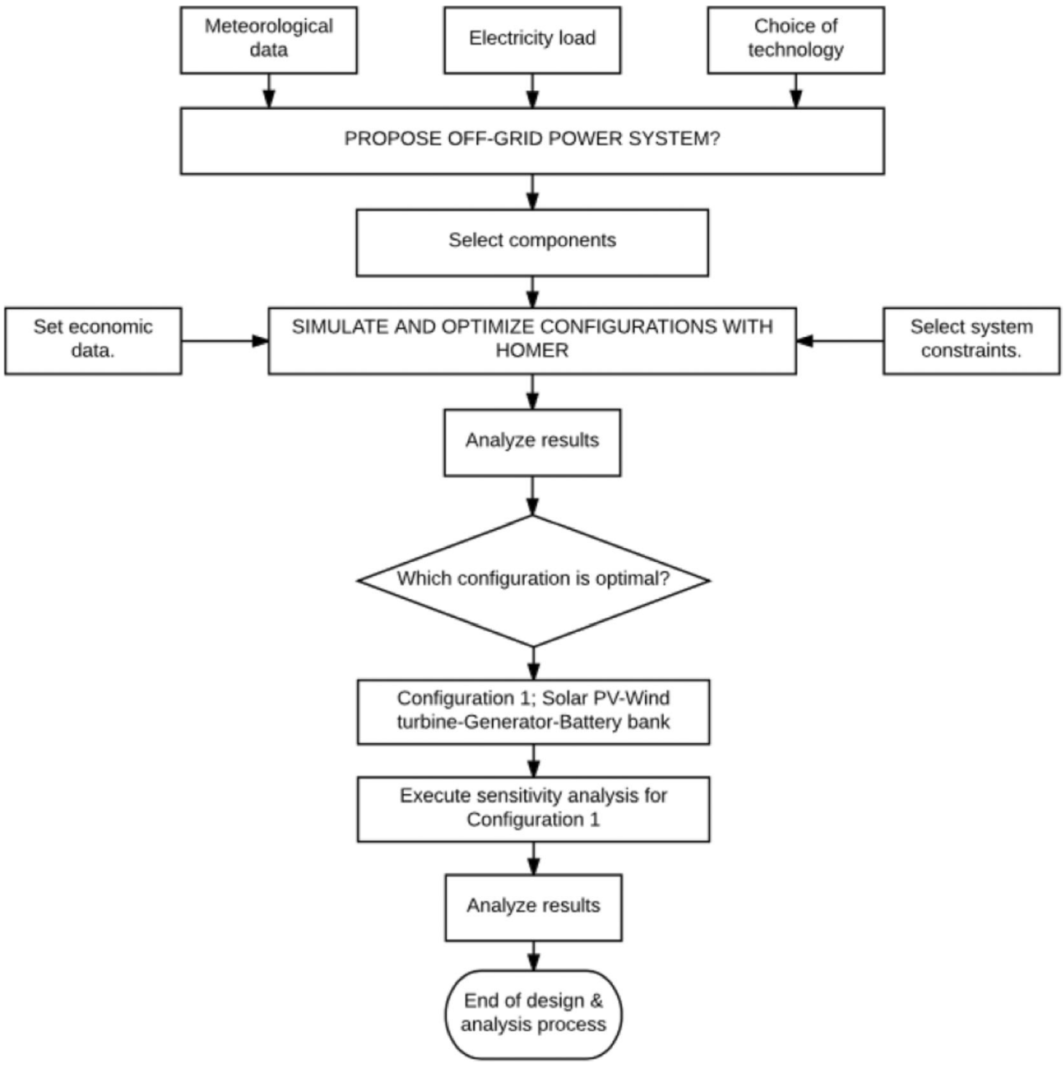

Fig. 1 Analysis framework of the design process

There is a strong demand for electricity at Korr as 86% of the 83 households surveyed are willing to be supplied with electricity. Of these, 46% are willing to pay upfront. Forty-five percent of the respondents use solar home systems as their main source of energy for lighting. Twenty-seven percent use battery-powered flashlights and 19% depend on kerosene.

Load profiles and energy resources

The demand for electricity in rural Kenya is not as high as in urban regions such as Nairobi. The basic energy demands in such rural sites can be classified as household, business, and institutional loads. Three different household load profiles were assumed (low, medium, and high income). Low-income customers were assumed to be 400, 320 of the medium income, and 80 of high income. There were also 2 more categories of load profiles taken into account: the business load profile (60 customers) and institutions, such as school, government institution, a medical center (17 customers). The assumed electricity consumption of the village will be used to calculate the total load, average daily load, peak load, and load factor. The results are summarized in Table 4 and Fig. 3.

Prior to implementing a renewable energy project, it is essential to assess the amount of resources available on the site or within a reasonable distance to the site. This is necessary to design an optimum solution that incorporates one or several renewable energy technologies. The village Korr is situated in the northern part of Kenya

Fig. 2 Arial view of Korr

where the terrain is rocky/sandy and natural energy resources are limited. The topography of the region means it has potential for solar and wind electricity production. All the solar resource information used for Korr were extracted from NASA Surface Meteorology. The NASA surface meteorology was also used to acquire wind data for the location. The wind speed at the location, based on a 10-year monthly average, does not exceed 5.09 m/s with a yearly average of 4.39 m/s (Fig. 4).

System modeling

The objective is design a power generation system that can fulfill the electricity load of the community with a reliable supply of electricity. That calls for a system that incorporates storage for flattening out supply irregularities and other components such as wind turbine(s) and PV panels for harnessing sources of renewable energy. The cost of purchasing, installing, and maintaining these components must be justifiable where the cost parameters, such as net present cost (NPC), cost of energy (COE), and initial investment cost are at their lowest.

For such a task, a computer software capable of modeling a defined hybrid power generation system and performing a cost-benefit analysis is necessary (Panagiotidis et al. 2018). According to Connolly et al. (2010) and Sinha and Chandel (2014) who conducted reviews of various energy system analysis software, HOMER is a widely recommended and proven software for such a task. It is a micro-grid software developed by

Table 4 Summary of load profiles

Electric load profile	Total daily load* (kWh)	Average load** (kW)	Peak load (kW)	Load factor***
Household	1447.6	60.3	230.8	0.26
Business	2650.8	110.5	240	0.46
Institution	1493.62	62.2	132.6	0.47
Total	5592.02			

*Total daily load = sum of all loads in 24 h

**Average load = total daily load/24 h

***Load factor = average load/peak load

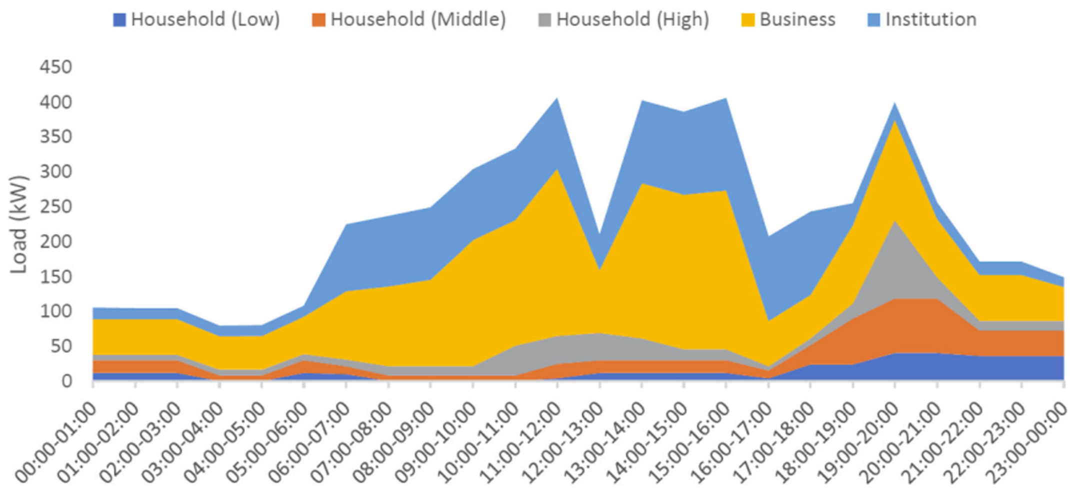

Fig. 3 Combined load profile for Korr village

the U.S National Renewable Energy Lab (NREL) in 1993, and offers its users three designated functions: simulation, optimization, and sensitivity analysis, with the ability to perform hourly time series analysis over a year for power generation systems of various energy inputs serving a specific load demand (Farret and Simões 2006). Simulation is the first function that models the performance of a power generation system to assess technical feasibility and life-cycle cost. A combination of wind turbines, hydro turbines, PV arrays and generators of any fuels, ac-dc converter, battery bank, electrolyzer, and hydrogen tank can devise the system configurations in an off-grid or on-grid mode (Table 5).

The second function of optimization explores system configuration that meets the technical constraints at the lowest total NPC and then ranking the feasible configurations accordingly. This process is based on user-input decision variables; it determines the mix of components in a power system along with the dispatch strategy and the size and quantity of each component. The third and final function of sensitivity analysis aids the user to determine the accuracy of simulation and optimization results or evaluate the trade-off and appropriateness of power systems under different presumed conditions.

The final power system may comprise of a wind turbine(s), photovoltaic panels, battery, and a diesel

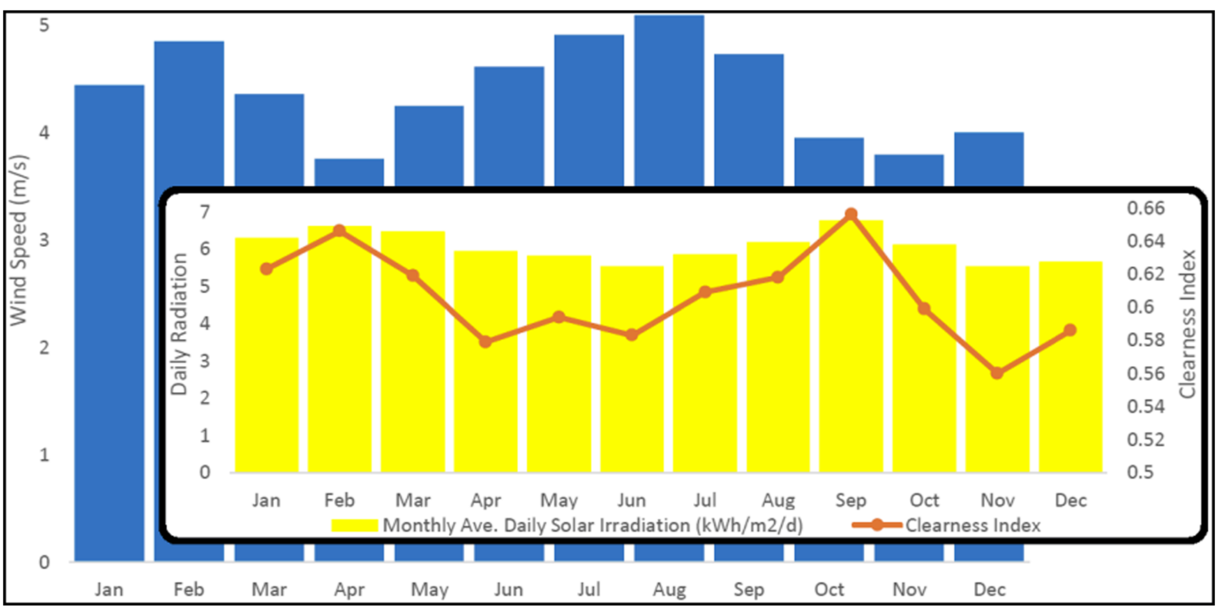

Fig. 4 Monthly average wind speed, daily solar irradiation, and clearness index

Table 5 Characteristics of PV array, wind turbine, diesel generator, battery, and inverter

	PV array	Wind	Generator	Battery	Inverter
Capital cost (US$)	US$1,200	US$687,500	US$450	US$1,150	US$425
Replacement cost (US$)	US$840	US$481,250	US$472.5	US$805	US$297
O & M cost (US$)	US$10	US$6,875	US$0.03	US$10	US$50
Lifetime (years)	25	25		18	15

generator. The decision variables for the power system accounts for the following:

- Size of wind turbine
- Size of PV array
- Battery autonomy
- Size of generator
- Size of converter
- Operation hour fraction of the diesel generator

Each specific component is modeled in the software HOMER, by entering its economic and performance properties. Although the charge controller is an important part of the system, it does not contribute to the simulation done with HOMER.

The PV power is simulated without sun tracking but the average monthly temperature of Korr is set in to HOMER, as it is known that the effect of temperature on the PV performances is considered during the simulation. The PV model was Sharp ND-250QCS (peak power: 250 W), of 15.3% efficiency and operating temperature of 47.5 °C.

A wind turbine with rated power of 275 kW (model: Vergnet GEV MP-C, hub hub height at 55 m) was selected for the design of the power system. The average installation cost for large wind turbines is used asserting that a kW of output costs US$2500. The operation and maintenance is assumed to be 1% of the capital costs.

A diesel generator was also selected for the simulation to provide backup power at times of high and unexpected demand or during times of lacking renewable energy resources. Running the engine in low, less than 30% of rated capacity power level, should be evaded as the engine will operate at a low efficiency, and it can potentially cause glazing on the cylinder walls. Diesel engines typically should run at 80% or above of its rated capacity to ensure the fuel consumption is at an optimum level. For the economic analysis, the average for the year 2017 was US$0.95/L.

For the simulation, the storage system utilized a lead-acid battery (model: Discover 2VREA-1500TF). The cost of the battery is chosen based on the average installed cost of batteries derived from (International Renewable Energy Agency 2016b) to be US$1150/battery while the replacement cost are assumed to be 70% of the installation costs. The annual operation and maintenance cost are assumed to be US$10 per battery as modern batteries require minimum maintenance. The nominal capacity of the battery is 1.48 kWh, the roundtrip efficiency is 85%, and the minimum state of charge 20%. Last, an inverter and a rectifier were needed for the system. The inverter and rectifier efficiencies were assumed to be 95 and 90% respectively. To avoid excessive discharging or overcharging of batteries, a charge controller was incorporated into the system. Overcharging can be dodged with either shunt or a series regulator (Markvart 2000).

In this study, certain constraints have been set for the power system. The maximum annual capacity shortage is set at 10% with a minimum renewable fraction of 30%. The operating reserve as a percentage of hourly load is set at 10%. The annual peak of 0% is with an operating reserve as a percentage of renewable output. Solar power output is set at 25% while wind power output is set at 50%.

Economic input

HOMER is equipped with an economic input window to successfully derive the NPC and other indicators of the system. The discount rate and the inflation rate have taken as 10 and 6.5%. This reflects the most up-to-date investment environment for Kenya (Anastasopoulou et al. 2016). Twenty-five years is the assumed lifetime of the off-grid system. In this time, the system can meet the load demand of the village that will remain constant.

Results and discussion

Optimization and proposed solutions

HOMER simulated 25,454 solutions where only 7019 were feasible. Twelve thousand six hundred forty-four were infeasible due to the capacity shortage and 5791 due to minimum renewable fraction. The feasible designed off-grid power generation systems are presented in this section. The details of the optimization results for the selected power system generating electricity for 800 households, 60 businesses, and 17 institutions were considered. The feasible solutions of the power systems were chosen and presented in an ascending order of the net present cost (NPC).

The authors proposed various configurations for further analysis and to increase the chance of finding the most optimized and capable system. Some criteria's that the setup must fulfill were having a low NPC, low COE, a higher renewable fraction, minimum fuel consumption, less capacity shortage, and smaller excess electricity. A system that can have all these factors checked off was considered as an optimum system. The following constraints were set up in HOMER to provide structure. They are as follows:

- A minimum renewable fraction of 30%
- Diesel price of US$0.95/L
- Maximum annual capacity shortage of 10%
- Primary load of 5592.2 kWh/day

All the feasible configurations under further analysis are displayed in Fig. 5. A further study of these factors is carried out to determine how each configuration performs under the named parameters. Furthermore, a simulation was run with the constraint of 0% renewable fraction to see how a diesel generator would compare to these systems:

1. Solar PV—wind turbine—generator—battery bank
2. Solar PV—generator-battery bank
3. Solar PV—wind turbine—battery bank
4. Solar PV—generator
5. Solar PV—wind turbine—generator
6. Wind turbine—generator—battery bank

The configuration of solar PV—wind turbine—generator—battery bank performs best with a NPC of US$10.8 million, an initial capital cost of US$4.26 million and an operating cost of US$388,747. This leaves it with a COE of US$0.314, the most favorable in terms of costs. This system is closely followed by the system with a configuration of solar PV—generator—battery bank with a COE of US$0.319. All systems have lower NPC and COE in comparison to the diesel generator; however, it is the system that has lowest initial capital costs. This proves that running a diesel generator is expensive and this can be backed by the fact that it has the highest operating costs (Fig. 6).

Figure 7I presents the renewable fraction of the different configurations. In other words, what percentage of the system is powered by renewable energy technology.

The system that does not incorporate a diesel generator which in this case is the configuration of solar PV—wind turbine—battery bank has a renewable fraction of a 100%, with no emissions produced. The two systems with a reasonable high renewable fraction of 74% solar PV—wind turbine—generator—battery bank and 59% solar PV—generator—battery bank are systems that also have the lowest COE. The fuel consumption of power systems is higher the lower the renewable fraction which comes as no surprise as the diesel generator must operate more hours to generate the power to meet the energy demands of the community. This results in the solar PV—wind turbine—generator—battery bank and the solar PV—generator—battery bank showing favorable numbers with the latter consuming 92,406 lt more annually. In the 25 years, the systems are assumed to operate this would be 2,310,150 lt of diesel that was not purchased and emitted into the atmosphere solar PV—wind turbine—generator—battery bank. Once again, the configuration incorporating all power generation technology is preferred.

Regarding excess electricity production (Fig. 7II), all systems can meet the energy demand of the village also exceeding it in all scenarios. Systems that produce too much excess electricity are perceived as costly and may be dimensioned incorrectly. The diesel generator is the system that performs the best in this parameter which may be down to the fact that it is not dependable on the weather like the other systems. The wind turbine—generator—battery bank, along with the solar PV—wind turbine—generator—battery bank and the solar PV—generator—battery bank are the other three systems that also perform somehow reasonable if compared to the other systems that incorporate renewable energy

	Architecture					
	Shar250 (kW)	Verg MP-C	Gen (kW)	2VRE4-1500TF	Converter (kW)	Dispatch
1	1,000	3	660	900	500	LF
2	1,250		660	600	500	LF
3	1,500	8		1,500	500	CC
4	1,500		660		500	CC
5	1,500	1	660		500	CC
6		7	660	900	300	LF

	Cost				System
	COE ($)	NPC ($)	Operating cost ($)	Initial capital ($)	Ren Frac (%)
1	$0.314	$10.8M	$388,747	$4.26M	74
2	$0.319	$11.0M	$490,448	$2.70M	59
3	$0.347	$11.1M	$244,077	$6.97M	100
4	$0.372	$12.8M	$622,827	$2.31M	36
5	$0.387	$13.3M	$612,770	$3.00M	38
6	$0.413	$14.2M	$584,532	$4.35M	45

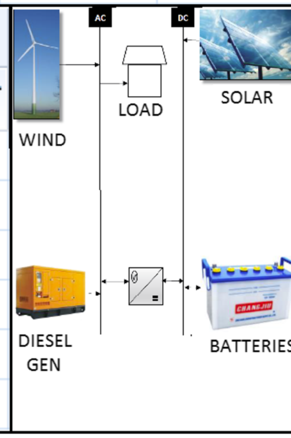

Fig. 5 Optimization results of categorized configurations and a schematic representation of potential system (bottom right corner)

technology. The three worst systems all lack a battery bank and a generator incorporated at the same time which maybe the reason behind their poor performance. The worst performing configuration of solar PV—wind

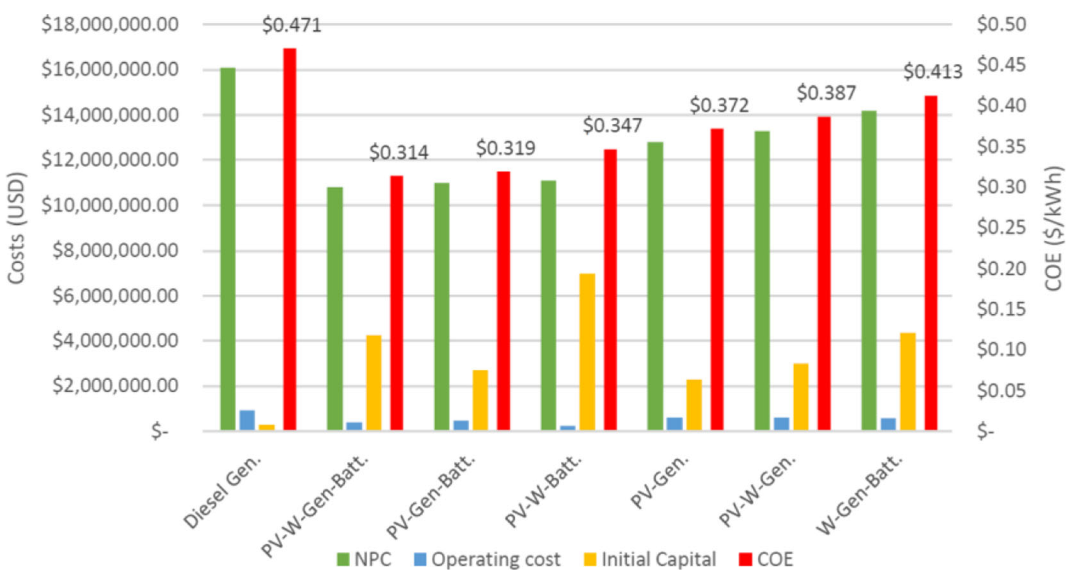

Fig. 6 Costs of the different configurations

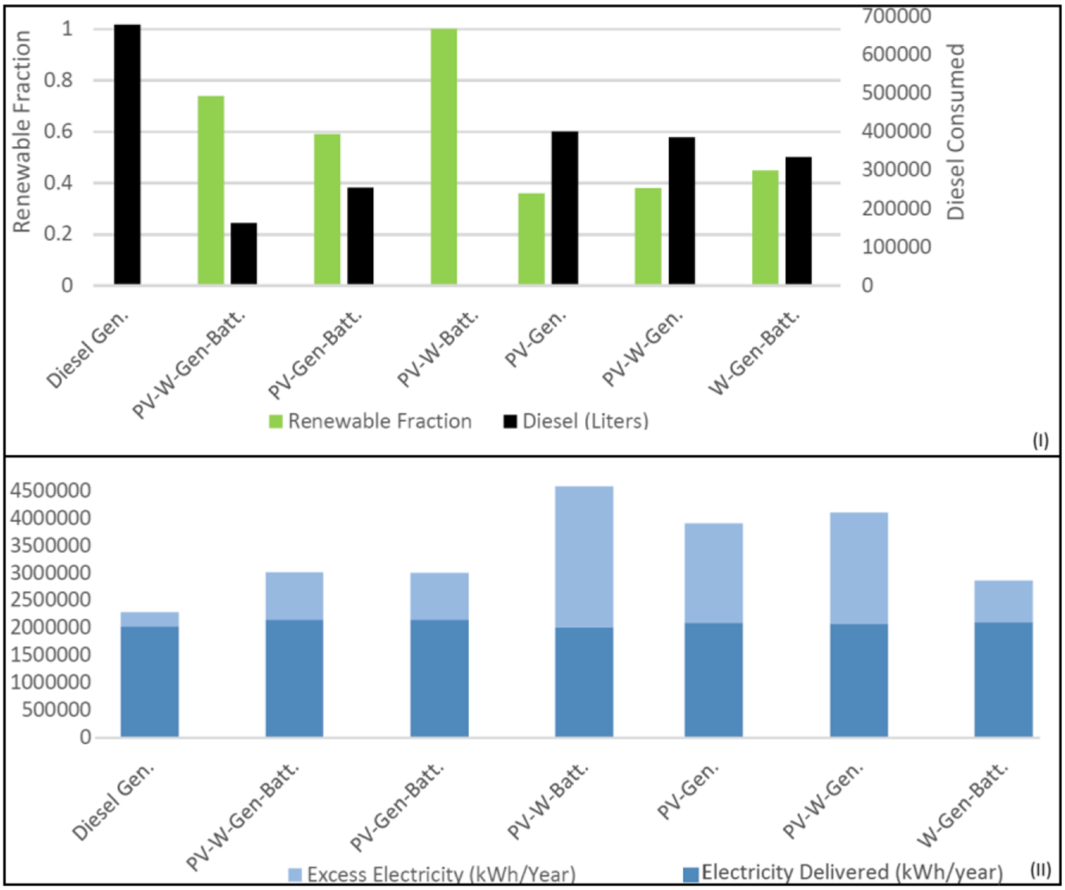

Fig. 7 Renewable fraction and diesel consumed of the configurations (I) and excess electricity produced by each configuration (II)

turbine—battery bank produces over 50% of excess electricity annually. Based on all the above and since Korr is 55 km away from the national grid that is currently under expansion and surrounded by other villages, this excess power could be sold to generate income. Therefore, the decision was made to select the configuration of solar PV—wind turbine—generator—battery bank as it performs satisfactory under the chosen parameters for analysis and does produce excessive electricity compared to the 100% renewable energy system. The diesel generator runs more often than would be preferred due to the load demand during the night hours. The solar PV system, the wind turbines, and the battery bank cannot meet this load demand without the diesel generator at the same COE. Based on the monthly analysis, there are two periods during the year when the electricity production by renewable energy technology drops slightly (Fig. 8). This may be explained by the two seasons of dry and rainy season Kenya experiences.

The capital cost of US$4.25 million makes up significant costs of the system as well as the fuel costs during the years of operation. Figure 9 shows the summary of the cash flow. The battery bank and the diesel generator stand out due to costs associated with. They account for more than 60% of the costs incurred for the system.

Table 6 illustrates the emissions from the optimized system as estimated by HOMER. If compared to a diesel system then the selected system is more environmental friendly and produces less pollutant across all categories. It is not possible to have wind and solar power as the sole source of electricity in a stable-base load due to its dependency on the weather. However, it does prove they can significantly reduce the use of conventional energy sources.

Sensitivity analysis

Sensitivity variables chosen by the authors will reveal how the optimum system is affected with the variation

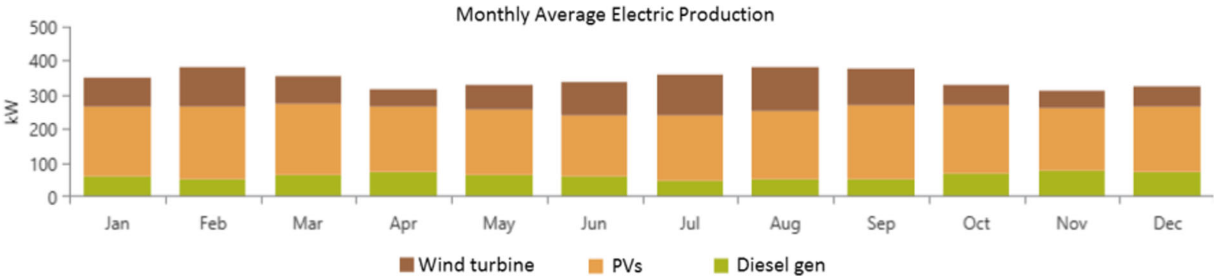

Fig. 8 Electricity generated per month based on power generation technology (extracted from HOMER)

of variables. The choice of these variables is driven behind that these have a huge impact on the cost-analysis of the system as the efficiency of the renewable technology depends hugely upon these parameters. The system will become more feasible when wind speed and solar radiation is higher and diesel price is lower and vice versa. The sensitivity variables considered and submitted in HOMER are presented in the Table 7.

The optimal system type graph is used to understand the optimization results. Figure 10I shows how sensitivity of the system varies in accordance with changes in wind speed and diesel price. In addition, the COE is also plotted in the graph as another indicator to show the system sensitivity (Fig. 10II). In this sensitivity analysis, the solar radiation was kept constant.

Discussion

The optimal system configuration obtained from the study shows that a cost-effective hybrid mini-grid can be implemented for decentralized electricity supply in Northern Kenya.

A question of financing the investment for the hybrid mini-grid arises on how to generate the large investment

Component	Capital ($)	Replacement ($)	O&M ($)	Fuel ($)	Salvage ($)	Total ($)
Autosize Genset Kenya	$297,000.00	$396,034.82	$780,966.26	$2,600,714.25	($10,256.08)	$4,064,459.25
Discover 2VRE4-1500TF JMH	$1,035,000.00	$1,886,412.85	$151,832.61	$0.00	($162,542.65)	$2,910,702.81
Sharp ND-250QCS JMH	$1,200,000.00	$0.00	$168,702.90	$0.00	$0.00	$1,368,702.90
System Converter	$212,500.00	$91,582.15	$421,757.25	$0.00	($22,093.21)	$703,746.19
Vergnet GEV MP-C [275kW]	$1,512,500.00	$0.00	$255,163.14	$0.00	$0.00	$1,767,663.14
System	$4,257,000.00	$2,374,029.82	$1,778,422.16	$2,600,714.25	($194,891.94)	$10,815,274.29

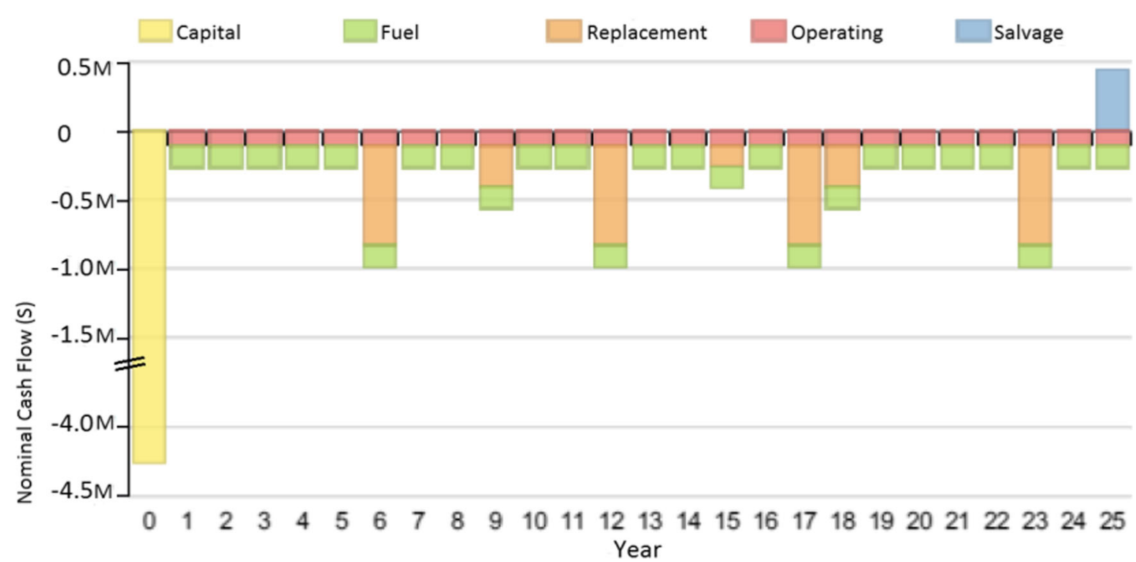

Fig. 9 Cash flow of net present cost of Components (US$) (extracted from HOMER)

Table 6 Emissions avoided

Quantity	PV-W-gen-batt (kg/year)	Diesel generator (kg/year)	Emissions avoided (kg/year)
Carbon dioxide	424,768	1,776,001	1,351,232
Carbon monoxide unburned	2677	11,194	8517
Hydrocarbons	116	488	371
Particulate matter	16	67	51
Sulfur dioxide	1040	4349	3308
Nitrogen oxides	2515	10,516	8001

volume of US$4,257,000. Most mini-grid developers rely on grants and subsidies for the early stage of the project. Grants may be provided by international development agencies, local government agencies or private individuals, and others. In the later stage of project development when risk is lower, projects could be financed through debt but must guarantee that the mini-grid will meet the revenue expectations.

To ensure the project implementation, an efficient revenue stream must be set up. The potential customers of Korr could be categorized on their needs and financial capability. A fixed weekly or monthly charge for usages can be applied for basic users while medium and large users can be connected with pre-pay or post-pay meters depending on their financial capacity. For the commercial aspects, such as bars or shops, a fixed charge can be applied or be fitted with meters on a "pay-as-you-go" basis.

The Kenyan government supports renewable energy technology, and therefore, the support from the government through feed in tariffs should be explored. This could significantly lower the COE for the users and make it on par with that supplied by the national grid but a more reliable source of power.

Regular interaction with the locals should be conducted to better understand the consumer needs but also educate them on the system for effective and productive use of electricity. Young individuals can be trained to ensure skilled labor supply for any future project expansion but also basic maintenance and operation. Through such undertakings, it is possible to empower the community which in turn may lead to the rise in demand for renewable mini-grids in isolated areas across the country.

Timely maintenance and replacement of components is important to ensure that reliable power is distributed to consumers but also to gain the community's confidence and loyalty. Furthermore, this could extend the operation of some of the components or even the system which could lead to a more profitable investment.

The results and findings will be discussed and evaluated in this section by reviewing the results and findings to evaluate the validity and usability of this study. This study had the objective of designing and analyzing an off-grid system capable of satisfying the energy demands of a village in Kenya.

The reliability of the study is expressed as the likelihood that another project would arrive at the same outcome by following the same research method. There are concerns on literature study and the collection of data that has been conducted through desk top research by applying the different websites available to the author.

A set of literature material stretching from scientific articles to books has been collected and analyzed. It is perceived to be of valid and reliable quality due to its scientific level. However, no systematic approach was established to choose the different literature as it was not possible to collect all the data on the subject. Such a concern could have been avoided by introducing a more well thought design for the search and selection process. The findings of the literature study form the foundation of this study and if other relevant literature has not been found it may influence the results.

Table 7 Sensitivity variables

Wind speed (m/s)	Diesel price (US$/lt)	Solar radiation (kWh/m^2/day)	Business and institutional load (kWh/day)
4.39, 3.64, 5.14	0.65, *0.95*, 1.25	5.32, *6.07*, 6.82	2702, 3108, *4144.6*

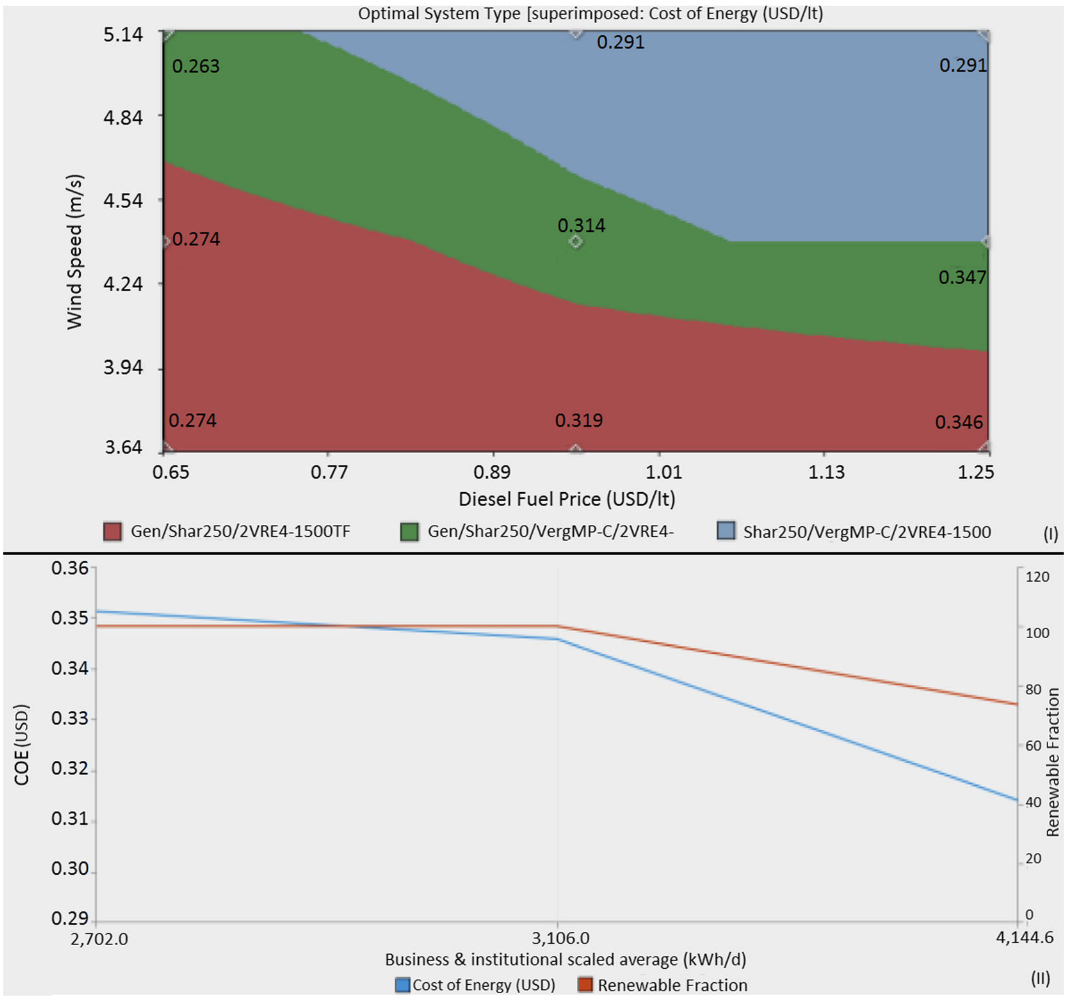

Fig. 10 Optimal system type (I) and primary (COE) and secondary (renewable fraction) variables (II)

Conclusions

The main objective of this study was to reveal the most cost-effective energy system to provide sufficient and reliable electricity through renewable and non-renewable resources if necessary for Korr, a village in Northern Kenya. After an analysis of different configurations using HOMER based on design constraints of low NPC, low COE, high renewable fraction, considerable excess electricity, and low fuel consumption. The configuration of solar PV—wind turbine—generator—battery bank is selected as the most cost-effective energy system. It can provide reliable power to the village with a renewable fraction of 74% where solar panels produce the majority of this electrical energy.

The projected COE is US$0.314/kWh with a NPC of US$10,815,274. The initial cost of implementing the hybrid mini-grid is estimated at US$4,257,000. The total lifespan of project is proposed to be 25 years with a salvage value of US$194,891.

Even though the cost of energy from the hybrid mini-grid is higher than that of the national grid, it is an ideal alternative for rural electrification. Currently, the implementation of a renewable energy system is not cost-competitive against conventional fossil fuel energy or grid connected power sources. However, in such cases, transferring “grid connectivity” to isolated networks is extremely expensive. It is no secret the need for clean power source and alternative solutions create good potential for widespread development of renewable energy systems. To add to this, the dependency on imported oil positions renewable energy systems as a promising energy compliment.

Hydroelectric power is the most dominant form of energy among all renewable energy technologies in Kenya. However, it is very dependable on high investment costs for dams and seasonal variation of rainfall. Therefore, effective hybrid mini-grids become a cost-effective to best complement for the limitation of hydropower and could be the solution that allows Kenya to accelerate its rural electrification rate and play its part in ensuring the 2030 goal for universal access for all becomes a reality.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

References

Alliance for Rural Electrification, 2014, Hybrid mini-grids for rural electrification - lessons learned, Available: https://www.ruralelec.org/publications/hybrid-mini-grids-rural-electrification-lessons-learned

Anastasopoulou, A., Butala, S., Patil, B., Suberu, J., Fregene, M., Lang, J., Wang, Q., & Hessel, V. (2016). Techno-economic feasibility study of renewable power systems for a small-scale plasma-assisted nitric acid plant in Africa. *Processes, 4*, 54. https://doi.org/10.3390/pr4040054.

Bekele, G., & Boneya, G. (2012). Design of a photovoltaic-wind hybrid power generation system for Ethiopian remote area. *Energy Procedia, 2011 2nd international conference on advances in energy engineering (ICAEE), 14*, 1760–1765. https://doi.org/10.1016/j.egypro.2011.12.1164.

Chauhan, A., & Saini, R. P. (2014). A review on integrated renewable energy system based power generation for stand-alone applications: configurations, storage options, sizing methodologies and control. *Renewable and Sustainable Energy Reviews, 38*, 99–120. https://doi.org/10.1016/j.rser.2014.05.079.

Chauhan, A., Saini, R. P. (2015). Renewable energy based off-grid rural electrification in Uttarakhand state of India: Technology options, modelling method, barriers and recommendations. *Renewable and Sustainable Energy Reviews, 51*(4581), 662–681.

Connolly, D., Lund, H., Mathiesen, B. V., & Leahy, M. (2010). A review of computer tools for analysing the integration of renewable energy into various energy systems. *Applied Energy, 87*, 1059–1082. https://doi.org/10.1016/j.apenergy.2009.09.026.

Conway, K.M., Lütken, S., Peña, J.O., Röser, F., Sharma, S., de Vit, C., 2013. Guidebook for the development of a nationally appropriate mitigation action on efficient lighting. United Nations Environment Programme.

Ebhota, W. S., & Inambao, F. L. (2017). Facilitating greater energy access in rural and remote areas of sub-Saharan Africa: small hydropower. *Energy and Environment, 28*(3), 316–329.

Energy Research Centre of the Netherlands, 2015. Kenya Market Assessment for Off-Grid Electrification, Available: https://www.renewableenergy.go.ke/asset_uplds/files/ERC%20IFC%20minigrids%20-%20final%20report%20-%20Final(1).pdf. Accessed 26 April 2017.

Farret, F.A., Simões, M.G., Integration of Alternative Sources of Energy, 2006, Integration of Alternative Sources of Energy.

Hills, J. M., Michalena, E., & Chalvatzis, K. J. (2018). Innovative technology in the Pacific: building resilience for vulnerable communities. *Technological Forecasting and Social Change, 129*, 16–26.

HOMER Energy, 2017. HOMER Pro, Available: https://users.homerenergy.com/. Accessed 26 April 2018.

IEA, 2016. World Energy Outlook 2016.

International Renewable Energy Agency, 2016a, Solar PV in Africa: costs and markets, Available: https://www.irena.org/DocumentDownloads/Publications/IRENA_Solar_PV_Costs_Africa_2016.pdf. Accessed 28 July 2017.

International Renewable Energy Agency, 2016b, Innovation outlook renewable mini-grids, Available: http://www.irena.org/DocumentDownloads/Publications/IRENA_Innovation_Outlook_Minigrids_2016.pdf. Accessed 28 July 2017.

Kenya Vision 2030, 2007. Popular version, Available: http://vision2030.go.ke/about-vision-2030/. Accessed 26 April 2018.

Kenya Power Lighting Company, 2016. KPLC 2016 Annual Report, Available: http://www.kplc.co.ke/content/item/2255/2016%2D%2D2017-full-annual-report-for-the-year-ended-30th-june-2017. Accessed 9 Aug 2017.

Khan, H.J., Huque, A.J., Hoque, M.S., Ahmed, K., Rahman, R., Ahmed, T., Alam, R., A snapshot of energy efficient off grid appliance market in Bangladesh, 2016, ICDRET 2016 - 4th international conference on the developments in renewable energy technology, 7421525.

Koroneos, C. J., Nanaki, E. A., & Xydis, G. A. (2012). Sustainability indicators for the use of resources – the exergy approach. *Sustainability, 4*, 1867–1878.

Lal, S., & Raturi, A. (2012). Techno-economic analysis of a hybrid mini-grid system for Fiji islands. *Int. J. Energy Environ. Eng., 3*, 10. https://doi.org/10.1186/2251-6832-3-10.

Malekpoor, H., Chalvatzis, K., Mishra, N., & Ramudhin, A. (2018). A hybrid approach of VIKOR and bi-objective integer linear programming for electrification planning in a disaster relief camp. *Annals of Operations Research*, 1–27.

Mandelli, S., Barbieri, J., Mereu, R., & Colombo, E. (2016). Off-grid systems for rural electrification in developing countries: definitions, classification and a comprehensive literature review. *Renewable and Sustainable Energy Reviews, 58*, 1621–1646. https://doi.org/10.1016/j.rser.2015.12.338.

Markvart, T., 2000. Solar electricity, 2nd ed. Wiley.

Michalitsakos, P., Mihet-Popa, L., & Xydis, G. (2017). A hybrid RES distributed generation system for Autonomous islands: a DER-CAM and storage-based economic and optimal dispatch analysis. *Sustainability, 9*(11), 2010. https://doi.org/10.3390/su9112010.

Mohammed, Y. S., Mustafa, M. W., & Bashir, N. (2014). Hybrid renewable energy systems for off-grid electric power: Review of substantial issues. *Renewable and Sustainable Energy Reviews, 35*, 527–539. https://doi.org/10.1016/j.rser.2014.04.022.

Nerini, F. F., Broad, O., Mentis, D., Welsch, M., Bazilian, M., & Howells, M. (2016). A cost comparison of technology approaches for improving access to electricity services. *Energy, 95*, 255–265. https://doi.org/10.1016/j.energy.2015.11.068.

Nema, P., & Dutta, S. (2012). Feasibility study of 1 MW standalone hybrid energy system: for technical institutes. *Low Carbon Economy, 3*, 63–68. https://doi.org/10.4236/lce.2012.33009.
Panagiotidis, P., Effraimis, A., & Xydis, G. A. (2018). An R-based forecasting approach for efficient demand response strategies in autonomous micro-grids. *Energy and Environment*, Article in Press. https://doi.org/10.1177/0958305X18787259.
Panagiotidou, M., Xydis, G., & Koroneos, C. (2016). Environmental siting framework for wind farms: a case study in the Dodecanese Islands. *Resources, 5*(3), 24. https://doi.org/10.3390/resources5030024.
Parajuli, R., Hussong, C., Ntoka, C., Charisoulis, G., Tulucan, T., & Sperling, K. (2015). Beyond oil and gas: possible future scenarios for the electricity sector in Saudi Arabia. *International Journal of Sustainable Energy, 34*(2), 71–92.
Rehman, S., & Al-Hadhrami, L. M. (2010). Study of a solar PV–diesel–battery hybrid power system for a remotely located population near Rafha, Saudi Arabia. *Energy, the 3rd international conference on sustainable energy and environmental protection, SEEP 2009, 35*, 4986–4995. https://doi.org/10.1016/j.energy.2010.08.025.
REN21, 2016. Renewables 2016 Global Status Report, Available: http://www.ren21.net/wp-content/uploads/2016/06/GSR_2016_Full_Report.pdf. Accessed 6 Dec 2017.
Sinha, S., & Chandel, S. S. (2014). Review of software tools for hybrid renewable energy systems. *Renewable and Sustainable Energy Reviews, 32*, 192–205. https://doi.org/10.1016/j.rser.2014.01.035.
The World Bank, 2014. Electric power consumption (kWh per capita).
Vieira, F. M., Moura, P. S., & de Almeida, A. T. (2017). Energy storage system for self-consumption of photovoltaic energy in residential zero energy buildings. *Renewable Energy, 103*, 308–320.
Weir, T. (2018). Renewable energy in the Pacific Islands: its role and status. *Renewable and Sustainable Energy Reviews, 94*, 762–771.
World Energy Outlook 2012, Organization for economic co-operation and development, Paris, 2012.